

Aditya Shukla

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Education

Vellore Institute of Technology

B.Tech in Computer Science (Specialization in AI/ML)

Bhopal, M.P.

2023–2027 | **8.52 CGPA**

Technical Skills

Languages: Python, SQL

Strong Subjects: Data Structures & Algorithms, DBMS, Operating Systems, OOP

Libraries & Frameworks: Flask, Scikit-learn, OpenCV, NumPy, Pandas, Ultralytics YOLO

Web & APIs: HTML, CSS, JavaScript, REST APIs

Tools: Git, GitHub, VS Code, Postman, AWS

Concepts: Machine Learning, Computer Vision, Feature Engineering, Model Evaluation

Experience

Machine Learning Intern

Apr 2026 – Jul 2026

Bodhganga Academy

Delhi (Hybrid)

- Developed ML models for personalized learning recommendations using user behavior and educational data.
- Built data preprocessing and feature engineering pipelines for efficient model training and evaluation.
- Integrated ML inference APIs with the web platform and optimized prediction performance through model monitoring.

Projects

DocuScale – Hybrid RAG Engine | *Python, FastAPI, Qdrant, FastEmbed, LiteLLM, Pytest*

- Engineered a modular, local-first Hybrid RAG system combining on-disk dense vector retrieval (Qdrant) and lexical search (BM25) with Reciprocal Rank Fusion (RRF) and cross-encoder rescoring.
- Designed an automated citation auditing layer enforcing strict deterministic source grounding ([doc#chunk]) and automated hallucination/out-of-domain refusal checks.
- Built a production-ready asynchronous FastAPI service delivering sub-second query endpoints and real-time Server-Sent Events (SSE) token streaming.
- Architected an offline, reproducible test suite with 99 pytest cases using mockable Protocols, achieving 100% pass rates without GPU, external network, or third-party API dependencies.

Bank Transaction ETL Pipeline | *Python, Pandas, SQL, DuckDB, YAML*

- Built an end-to-end ETL pipeline for synthetic bank transactions, covering data extraction, transformation, validation, warehouse loading, and analytics reporting.
- Implemented configurable data quality checks to validate transaction amount, channel, status, and account fields, routing invalid records to a rejected dataset with explicit rejection reasons.
- Developed a CLI-driven pipeline with YAML configuration and incremental processing support, enabling reproducible ETL runs, validation checks, and automated analytics report generation.

Lane Detection & Driver Assistance System | *Python, OpenCV, NumPy, Computer Vision*

- Developed a real-time lane detection system using OpenCV and NumPy for processing image and video streams.
- Implemented edge detection, HLS color masking, and Region of Interest (ROI) filtering to accurately identify lane boundaries under varying road conditions.
- Integrated Optical Flow and Harris Corner Detection techniques to improve lane tracking stability and motion estimation across consecutive frames.
- Designed a lane departure warning system with confidence scoring and real-time visual overlays to enhance driving assistance.

Certifications

Applied Machine Learning in Python

University of Michigan (Coursera)

Cloud Computing

NPTEL (IIT Kharagpur)

Introduction to Internet of Things

NPTEL (IIT Kharagpur)

Student Innovation Workshop

T-Hub, Hyderabad